Shaurya Khandelwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.60
Aklank Public School, Basant Vihar, Kota, Rajasthan	Class XII (CBSE)	2024	94.60%
Amity International School, Vasundhara, Ghaziabad, Uttar Pradesh	Class X (CBSE)	2022	98.60%

SCHOLASTIC ACHIEVEMENTS

National 3rd Rank in **e-Yantra Robotics Competition (eYRC 2025-26)** organized by **IIT Bombay**; competed in the *Holo Battalion* theme against **400+ teams** nationwide in a rigorous 7-month engineering challenge.

PROJECTS

Coordinated Multi-Agent Warehouse Automation (Holo Battalion)

IIT Bombay (e-Yantra Robotics Competition) | *Aug 2025 - Mar 2026*

- **Role**: Lead Cyber-Physical Systems Developer (Chassis design to multi-agent software deployment)
- **Tech Stack**: ROS 2 Humble, Python, OpenCV, MQTT, ESP32, Gazebo Fortress, Fusion 360, Linux.
- **Fleet Coordination & FSM**: Designed and deployed a centralized Finite State Machine (FSM) in ROS 2 that coordinated **3 physical holonomic robots** executing parallel sorting and stacking tasks.
- **Advanced Path Planning**: Developed a *Cooperative Space-Time A* planner* operating with a time-step reservation table, preventing deadlocks and inter-robot collisions in shared workspace corridors.
- **Perception & Coordinate Warping**: Implemented an overhead camera localization pipeline using **OpenCV** and ArUco markers; computed a **homography warping matrix** for pixel-to-world conversion and derived height-offset scale correction formulas to eliminate 3D perspective projection errors.
- **Control Systems & Kinematics**: Derived the inverse kinematics equations for a 3-wheel omni-directional drive system; programmed closed-loop **PID controllers** on *x*,*y*, axes with **dead-band compensation** to achieve a <10mm tracking tolerance.
- **CAD, PCB & Manipulation**: Modeled the custom robot chassis using **Autodesk Fusion 360** and wired physical circuit schematics; integrated jointed servo arms with MOSFET-driven electromagnets to autonomously construct stable **2-level vertical stacks** of physical crates.

SketchParty -- Real-Time Multiplayer Drawing & Guessing Game

Personal Project | *Jun 2026 - Aug 2026*

- **Role:** Full-Stack Engineer -- canvas rendering engine, real-time WebSocket architecture, server-side game state machine, and cloud deployment
- **Tech Stack:** React, Vite, Node.js, Express.js, Socket.io, HTML5 Canvas API, Zustand | Deployed on Railway (server) + Vercel (client)
- **Real-Time Sync Engine:** Architected a vector-based stroke-streaming pipeline using Socket.io; accumulated pointer events into **16ms batched payloads**, reducing socket traffic **~10×** and achieving **sub-50ms stroke latency** across **8 concurrent clients**.
- **Server-Authoritative Game Loop:** Engineered a **5-state FSM** (Lobby -> Countdown -> Drawing -> Round End -> Game Over) on a Node.js server; server owns all state transitions, rotating drawer selection, and time-scaled scoring -- eliminating race conditions on simultaneous correct guesses.

SKILLS

- **Full-Stack & Real-Time Systems**: React.js, Node.js, Express.js, Socket.io, WebSockets, HTML5 Canvas API, Zustand, Vite.
- **Programming Languages & CS Fundamentals:** Python, C++, C, JavaScript, HTML, CSS, Data Structures & Algorithms, MATLAB, Bash/Shell Scripting.

- **Robotics Middleware & Simulation**: ROS 2 (Humble), Gazebo Fortress.
- **Control & Navigation:** Space-Time A* Path Planning, Inverse Kinematics, Closed-Loop PID Control.
- **Version Control & Collaboration:** Git, GitHub, CI/CD (Railway, Vercel)
- **Perception & Computer Vision:** OpenCV, Camera Calibration, Homography Warping, ArUco Tracking
- **CAD & Hardware Prototyping:** Autodesk Fusion 360 (CAD Modeling), EasyEDA (Circuit Design), 3D Printing.
- **Embedded Systems & IoT:** ESP32, MQTT Protocol, PWM Motor Control, MOSFET Drivers.

POSITIONS OF RESPONSIBILITY

Core Member at **Robotix (The Robotics Club of IIT Hyderabad)**

Core Member at **Gesture (The Art Club of IIT Hyderabad)**

RELEVANT COURSES

Introduction to Programming, Automation Lab, Data Structures and Applications, Introduction to Probability, Introduction to Statistics